

Lab 4

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```
source("http://thegrantlab.org/misc/cdc.R")
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

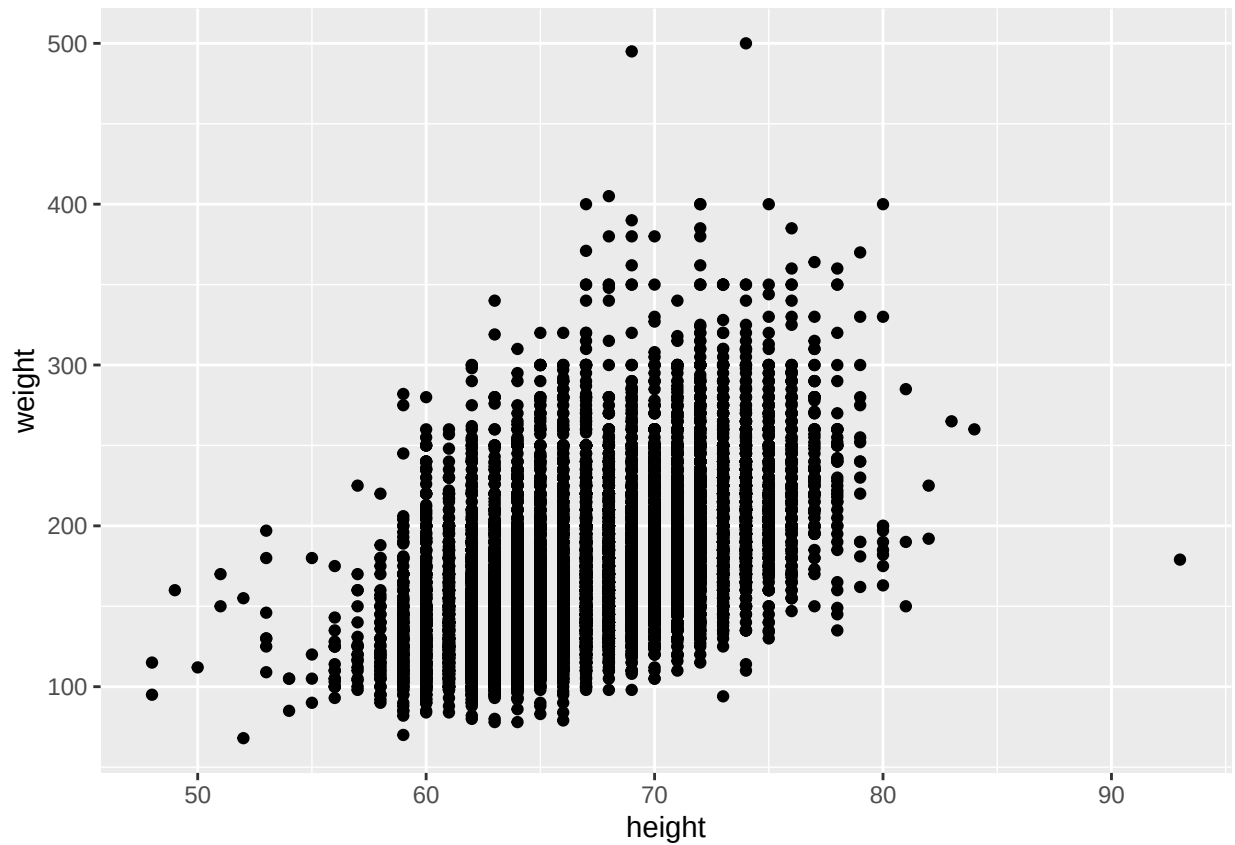
## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(ggplot2)
tail(cdc, 10)
```

```
##      genhlth exerany hlthplan smoke100 height weight wtdesired age gender
## 19991 excellent      1        1         0     71    195      190  43      m
## 19992 very good      1        1         1     72    210      175  52      m
## 19993 very good      1        1         0     71    180      180  36      m
## 19994 very good      0        1         1     63    165      120  31      f
## 19995      good      0        1         1     69    224      224  73      m
## 19996      good      1        1         0     66    215      140  23      f
## 19997 excellent      0        1         0     73    200      185  35      m
## 19998      poor      0        1         0     65    216      150  57      f
## 19999      good      1        1         0     67    165      165  81      f
## 20000      good      1        1         1     69    170      165  83      m
```

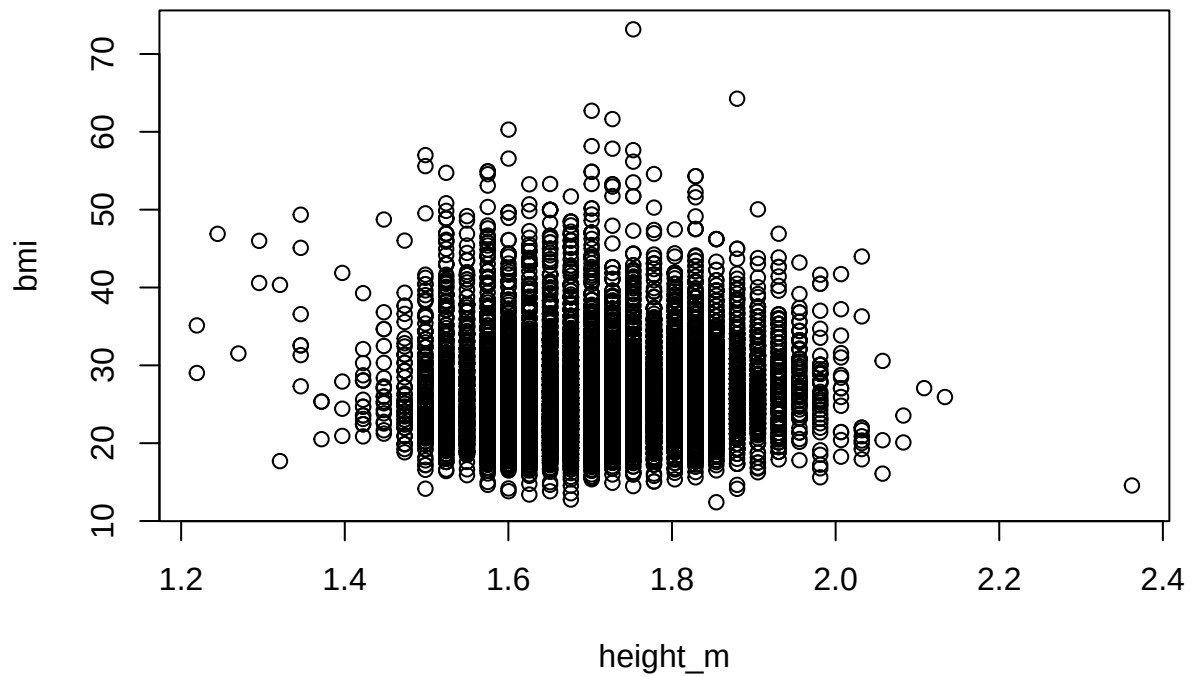
```
## Plots
ggplot(cdc, aes(height, weight)) + geom_point()
```



```
cor(cdc$height, cdc$weight)
```

```
## [1] 0.5553222
```

```
weight_kg <- cdc$weight * 0.454  
height_m <- cdc$height * 0.0254  
bmi <- weight_kg / (height_m^2)  
plot(height_m, bmi)
```



```
cor(height_m, bmi)
```

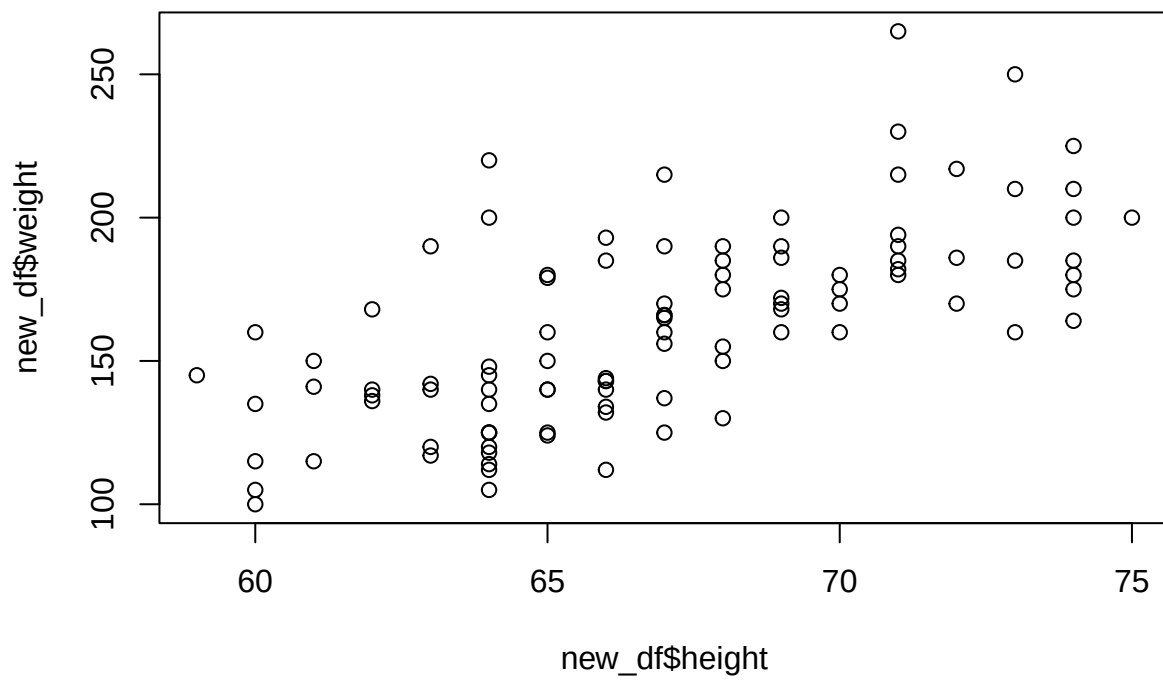
```
## [1] 0.03251694
```

```
sum(bmi>=30)
```

```
## [1] 3897
```

```
new_df <- cdc[1:100,]
```

```
plot(new_df$height, new_df$weight)
```



```
sum(cdc[bmi>=30, 9]=="m")
```

```
## [1] 1961
```